Sri Lanka Institute of Information Technology

IT3021 - DWBI - Assignment 1

IT23203662 – AYYASH M R Y

Y3S2.WE.DS.01.02

01. Data Set Selection

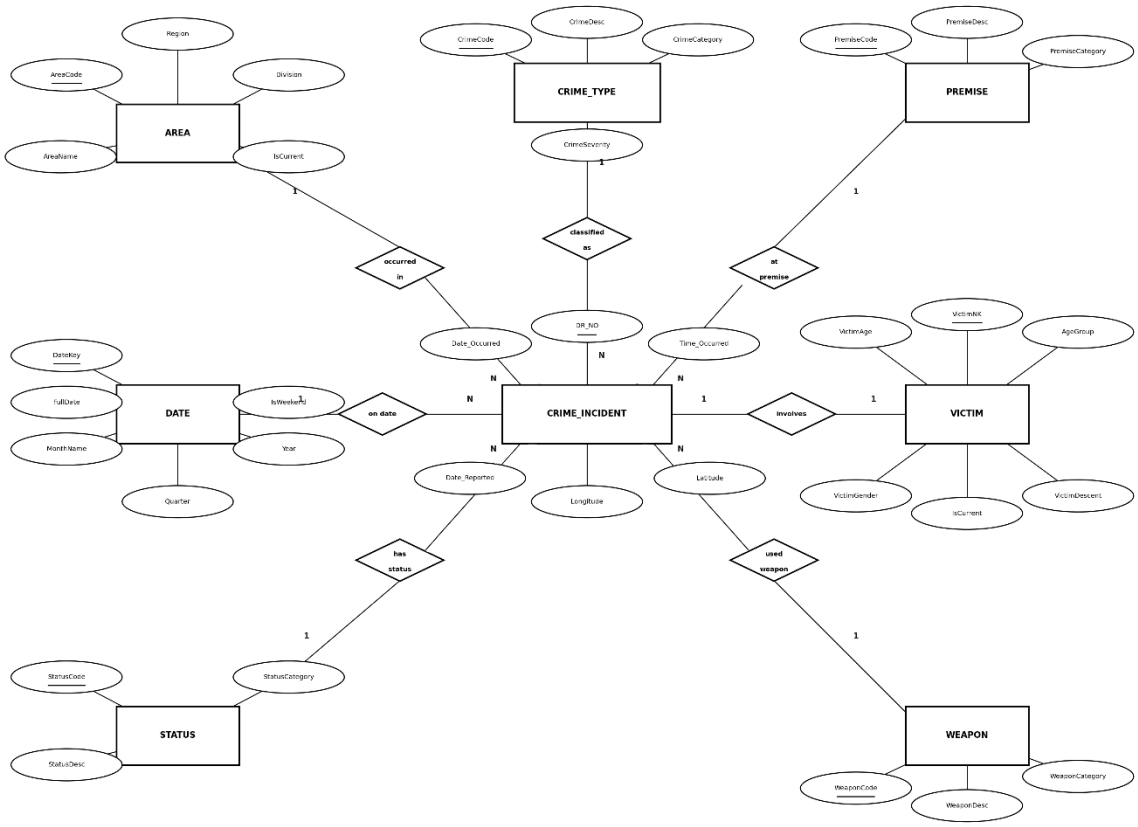
Dataset: LAPD Crime Incidents

Data Source: Los Angeles Police Department (LAPD) Open Data Portal

The LAPD Crime Incidents dataset contains detailed information on crime incidents reported in the City of Los Angeles from 2019 to 2025. It includes data on crime types, locations, victim demographics, weapons used, case status, and the reporting area (LAPD division). This dataset is suitable for crime trend analysis, geographic hotspot mapping, victim demographic analysis, and case resolution rate reporting.

The dataset was obtained from the LAPD Open Data Portal in CSV format and enriched with additional lookup reference files to support dimensional modelling. With over 50,000 records and multiple coded attributes (crime codes, area codes, weapon codes, premise codes), it provides sufficient data to demonstrate all key data warehousing concepts including Star Schema design, Slowly Changing Dimensions, ETL development, and accumulating fact tables.

LAPD Crime Analysis — Entity Relationship Diagram (Chen Notation)



02. Preparation of Data Sources

All data sources were obtained in CSV format or Excel format. During the data preparation phase, the original files were reviewed, and additional derived attributes were introduced to enrich the data for dimensional modelling purposes. The files were either loaded directly into SQL Server staging tables or used as flat file sources in SSIS.

Final Format of Source Data Before Transformation:

1. CSV Files (.csv)

- CrimeIncidents.csv – main crime incident records (50,000 rows)
- AreaCodeDescription.csv – LAPD area/division code lookup
- CrimeCodeDescription.csv – crime type code lookup
- PremiseCodeDescription.csv – premise/location type code lookup
- StatucCodeDescription.csv – case status code lookup
- WeaponCodeDescription.csv – weapon type code lookup
- AccumulatingFactUpdate.csv – case completion timestamps for Task 6

2. Excel File (.xlsx)

- VictimData.xlsx – victim demographic data (1,004,681 rows) containing Digital Record Number, Victim Age, Victim Gender, and Victim Descent code

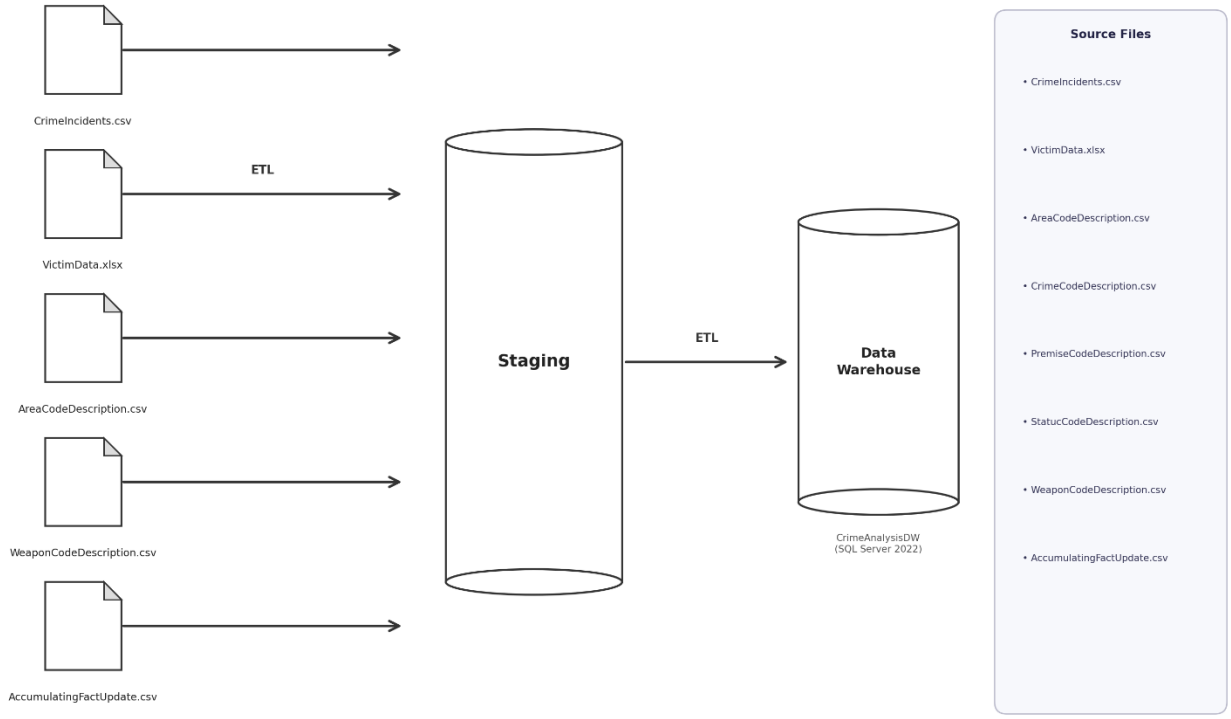
The CSV files were loaded into SQL Server staging tables (STG_*) via SSIS Flat File Source components. The Excel file was loaded using the SSIS Excel Source component, demonstrating the use of two different data source types as required.

AccumulatingFactUpdate	26/03/2026 01:38	Comma Separated...	147 KB
AreaCodeDescription	26/01/2025 18:47	Comma Separated...	1 KB
CrimeCodeDescription	26/01/2025 18:47	Comma Separated...	5 KB
CrimeDataNormalized	25/03/2026 03:14	Comma Separated...	151,939 KB
CrimeIncidents	26/03/2026 01:38	Comma Separated...	4,415 KB
PremiseCodeDescription	26/01/2025 18:47	Comma Separated...	10 KB
StatucCodeDescription	26/01/2025 18:47	Comma Separated...	1 KB
VictimData	25/03/2026 23:23	Comma Separated...	17,639 KB
WeaponCodeDescription	26/01/2025 18:47	Comma Separated...	2 KB
VictimData	26/03/2026 01:04	Microsoft Excel W...	25,173 KB

2.1 Data Source Summary

1	CrimeIncidents.csv	CSV	50,000	Main fact data – crime incident records
2	VictimData.xlsx	Excel	1,004,681	Victim demographics – feeds DIM_VICTIM
3	AreaCodeDescription.csv	CSV	22	Area lookup – feeds DIM_AREA
4	CrimeCodeDescription.csv	CSV	141	Crime type lookup – feeds DIM_CRIME
5	PremiseCodeDescription.csv	CSV	316	Premise lookup – feeds DIM_PREMISE
6	StatucCodeDescription.csv	CSV	8	Status lookup – feeds DIM_STATUS
7	WeaponCodeDescription.csv	CSV	81	Weapon lookup – feeds DIM_WEAPON
8	AccumulatingFactUpdate.csv	CSV	5,000	Case completion times – Task 6

03. Solution Architecture



CrimeAnalysisDW Staging Tables:

- STG_CrimeIncidents
- STG_VictimData
- STG_AccumulatingUpdate
- STG_CrimeCode
- STG_PremiseCode
- STG_WeaponCode
- STG_AreaCode

CrimeAnalysisDW Data Warehouse Tables:

- DIM_DATE
- DIM_TIME
- DIM_AREA (SCD Type 2)
- DIM_CRIME
- DIM_PREMISE
- DIM_WEAPON
- DIM_STATUS
- DIM_VICTIM (SCD Type 2)
- FACT_CRIME_INCIDENT

Architecture Components:

Data Sources

- CSV flat files containing crime incident records and dimension lookup data
- Excel file containing victim demographic records
- Generated CSV file for accumulating fact updates

Extract, Transform and Load

- Extraction phase – Retrieving data from CSV and Excel source files into staging tables
- Transformation phase – Deriving category columns, resolving surrogate keys, applying SCD Type 2 logic
- Loading phase – Inserting transformed data into dimension and fact tables

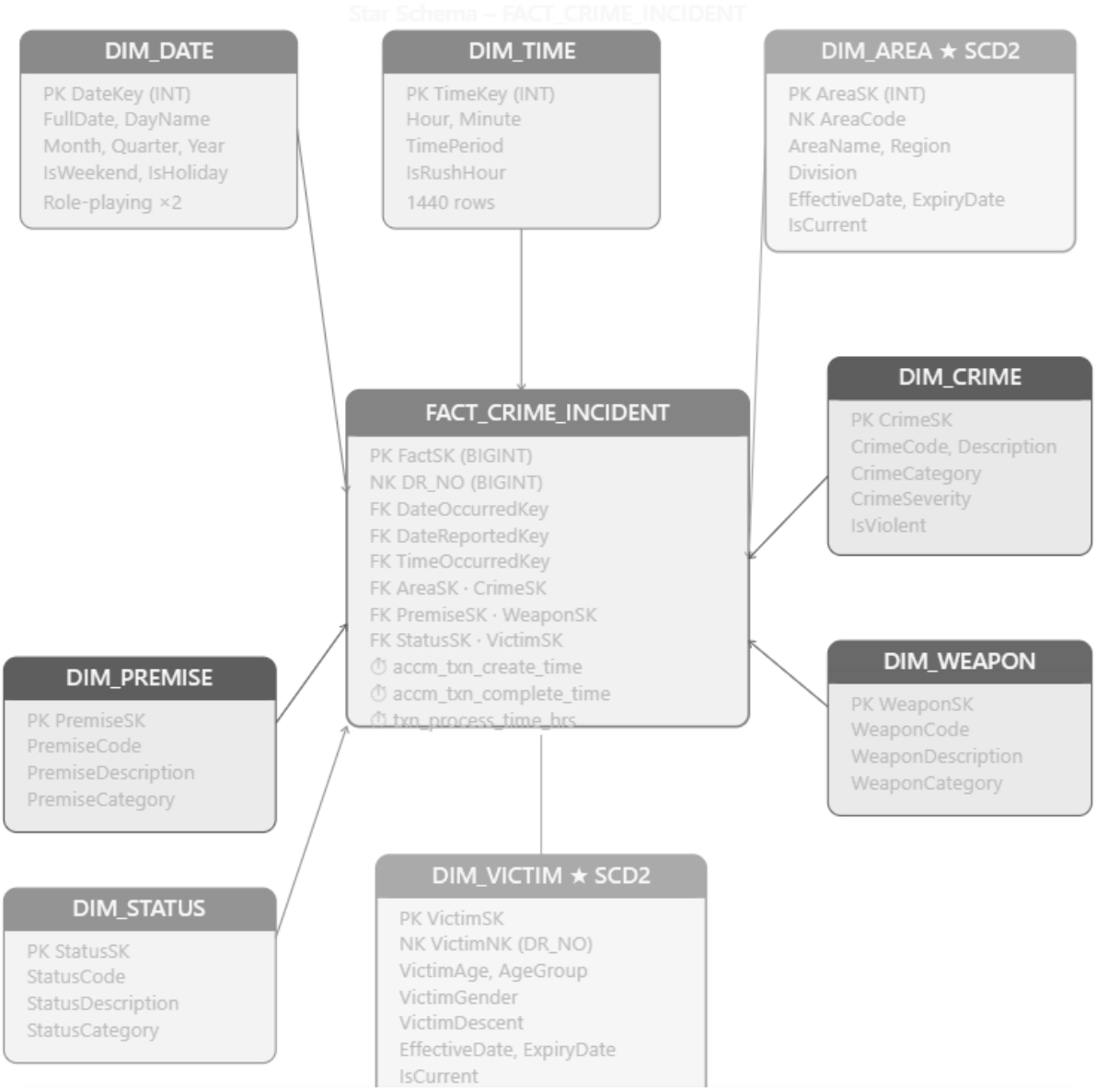
Data Warehouse

- Star Schema dimensional model with one central fact table and seven dimension tables
- Dimensional modelling using Facts and Dimensions
- Schema approach – Star Schema with surrogate keys and foreign key relationships

SSIS Project Structure

04. Data Warehouse Design & Development

Relational Diagram – Star Schema



This schema consists of seven-dimensional tables and one fact table.

- DIM_AREA implements Slowly Changing Dimension Type 2 because LAPD area names and regional classifications can change over time.
- DIM_VICTIM implements Slowly Changing Dimension Type 2 because victim demographic records may be corrected after initial recording.
- The grain of the fact table is one row per crime incident report (DR_NO).
- DIM_DATE is pre-populated for 2010–2027 and plays two roles: DateOccurred and DateReported.

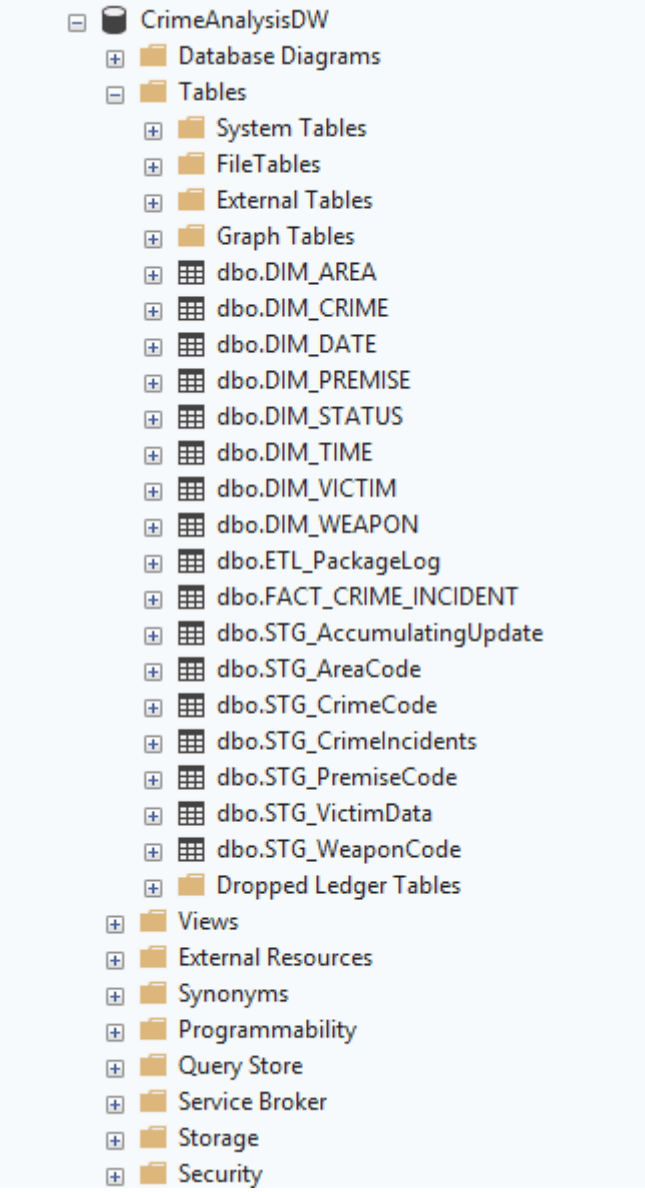
4.1 Fact Table – FACT_CRIME_INCIDENT

FactSK	BIGINT IDENTITY PK	Auto-generated surrogate primary key
DR_NO	BIGINT NK	Unique crime report number (natural key)
DateOccurredKey	INT FK → DIM_DATE	Date the incident occurred
DateReportedKey	INT FK → DIM_DATE	Date the incident was reported
TimeOccurredKey	INT FK → DIM_TIME	Time of occurrence (HHMM)
AreaSK	INT FK → DIM_AREA	LAPD division (SCD2 current row)
CrimeSK	INT FK → DIM_CRIME	Crime type
PremiseSK	INT FK → DIM_PREMISE	Premise/location type
WeaponSK	INT FK → DIM_WEAPON	Weapon used
StatusSK	INT FK → DIM_STATUS	Case investigation status
VictimSK	INT FK → DIM_VICTIM	Victim demographics (SCD2 current)
IncidentCount	INT	Always 1 – used in COUNT aggregations
DaysToReport	INT	Days between occurrence and reporting
VictimAge	INT	Age of victim at time of incident
Latitude / Longitude	DECIMAL(9,6)	Geographic coordinates
accm_txn_create_time	DATETIME	Set to GETDATE() at initial load
accm_txn_complete_time	DATETIME	Updated by Package 2 when case completes
txn_process_time_hours	Computed PERSISTED	Auto-calculated processing time in hours

4.2 Slowly Changing Dimensions

EffectiveDate	The date from which this version of the row is valid
ExpiryDate	NULL for the current record. Set to yesterday when a new version is inserted.
IsCurrent	1 = current/active row. 0 = historical/expired row.

4.3 Database Tables in SSMS



4.4 Row Counts Verification

```sql
USE CrimeAnalysisDW;

SELECT 'FACT_CRIME_INCIDENT' AS TableName, COUNT(*) AS [RowCount] FROM FACT_CRIME_INCIDENT
UNION ALL SELECT 'DIM_DATE',     COUNT(*) FROM DIM_DATE
UNION ALL SELECT 'DIM_TIME',     COUNT(*) FROM DIM_TIME
UNION ALL SELECT 'DIM_AREA',     COUNT(*) FROM DIM_AREA
UNION ALL SELECT 'DIM_CRIME',    COUNT(*) FROM DIM_CRIME
UNION ALL SELECT 'DIM_PREMISE',  COUNT(*) FROM DIM_PREMISE
UNION ALL SELECT 'DIM_WEAPON',   COUNT(*) FROM DIM_WEAPON
UNION ALL SELECT 'DIM_STATUS',   COUNT(*) FROM DIM_STATUS
UNION ALL SELECT 'DIM_VICTIM',   COUNT(*) FROM DIM_VICTIM;
```

100 % 24 0

Results Messages

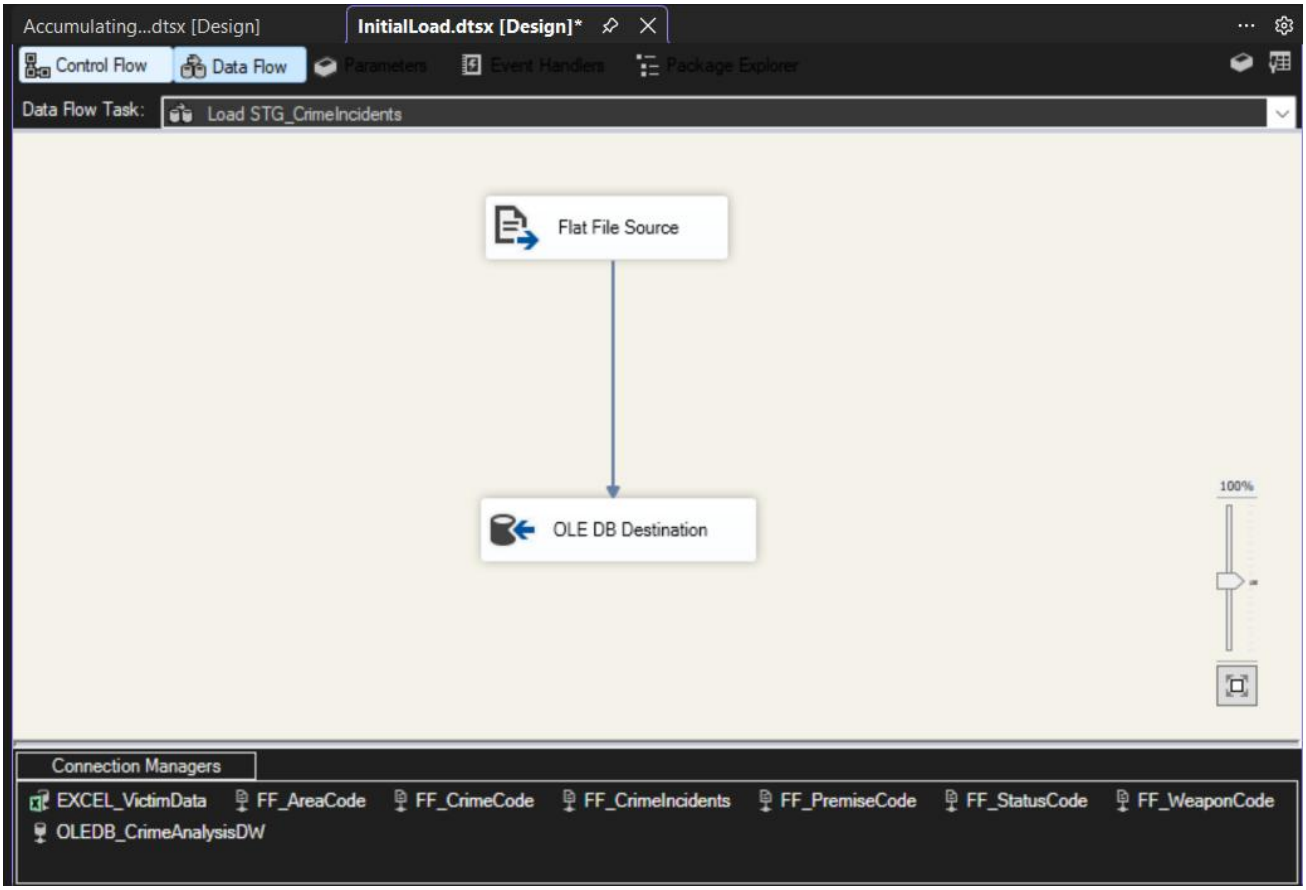
	TableName	RowCount
1	FACT_CRIME_INCIDENT	50000
2	DIM_DATE	6574
3	DIM_TIME	1440
4	DIM_AREA	43
5	DIM_CRIME	140
6	DIM_PREMISE	315
7	DIM_WEAPON	80
8	DIM_STATUS	16
9	DIM_VICTIM	1

05. ETL Development

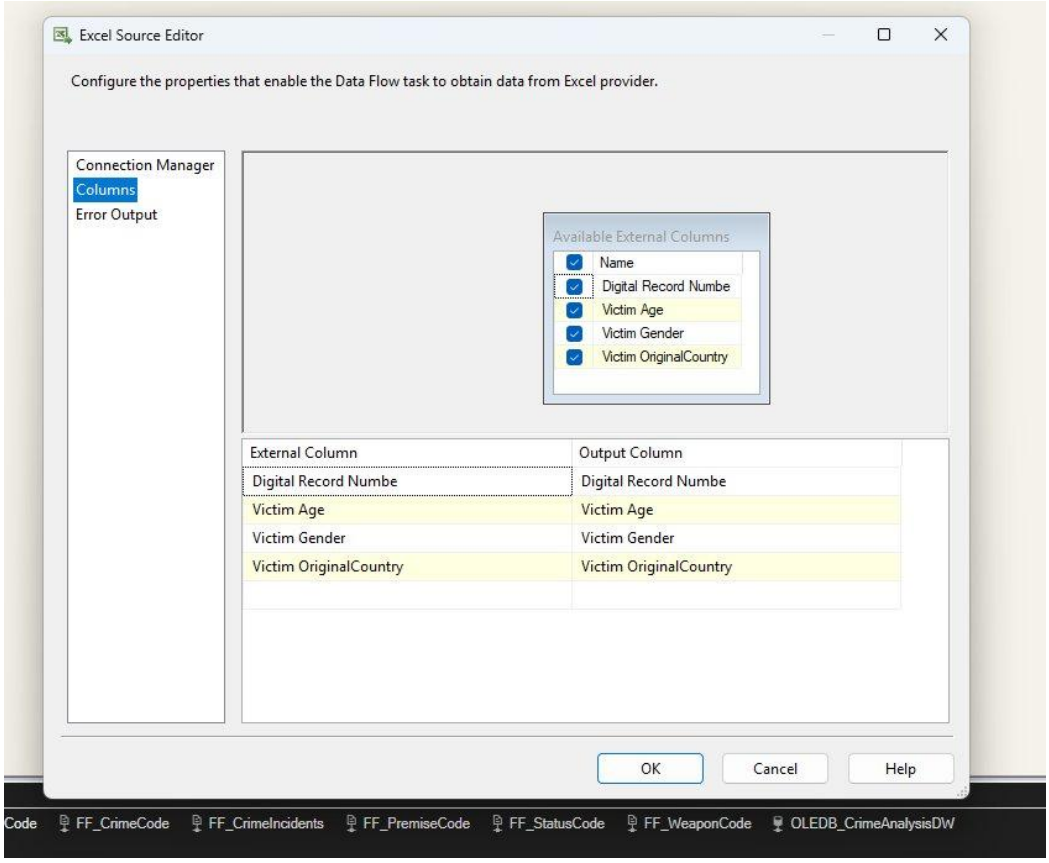
5.1 Staging Tables

Load the data from source files to staging tables. This is done using SSIS Data Flow tasks with Flat File Source and Excel Source components connected to OLE DB Destination components targeting the staging tables.

5.1.1 Load STG_CrimeIncidents from CSV

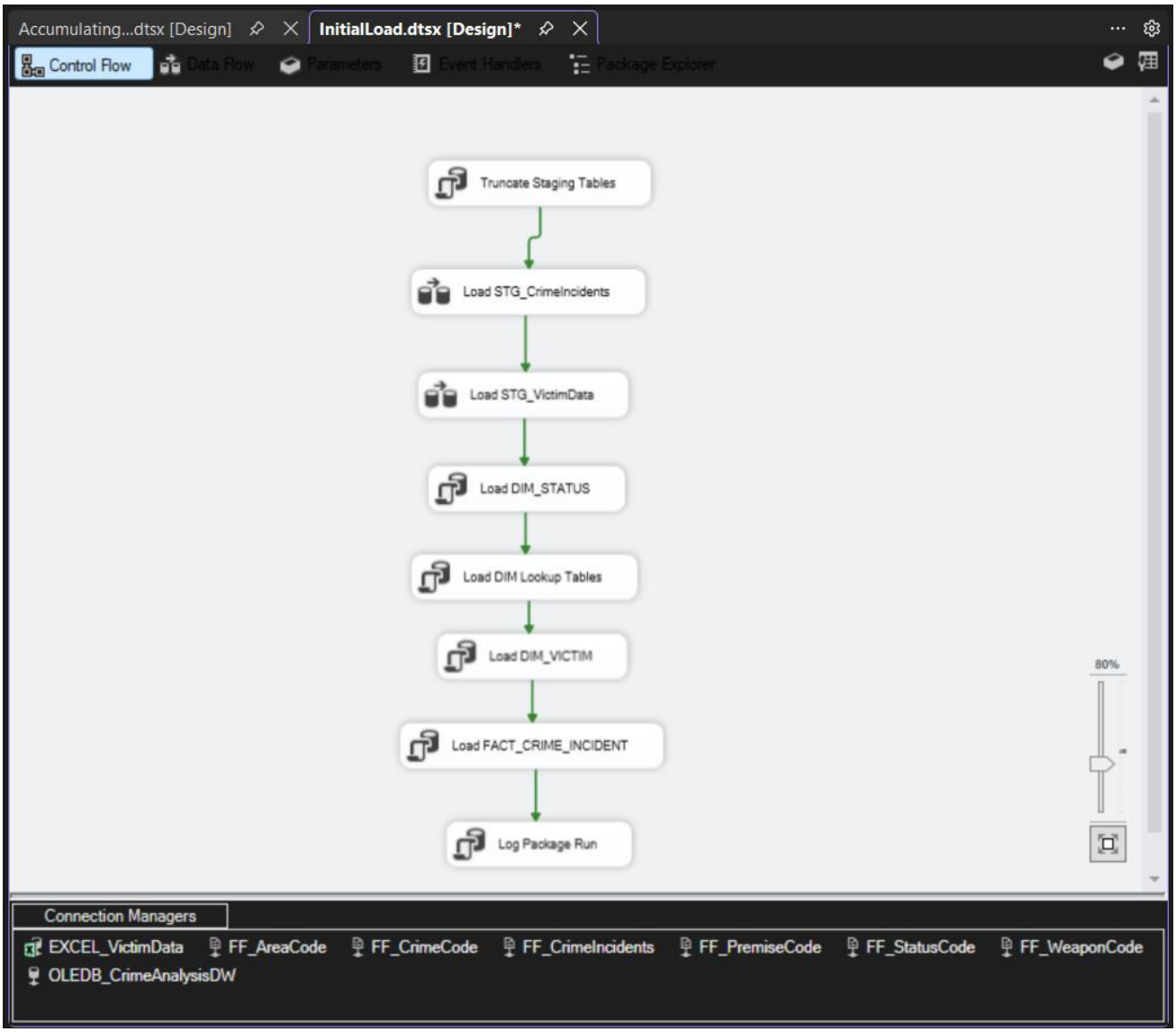


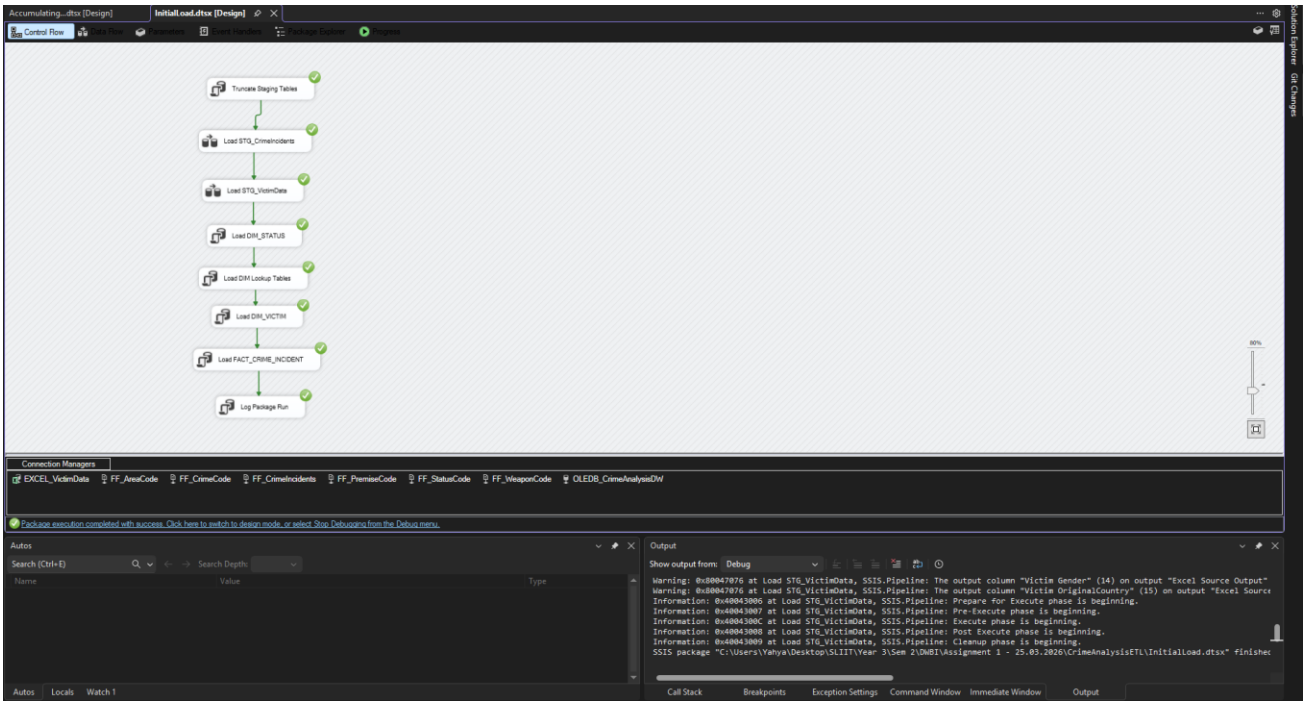
5.1.2 Load STG_VictimData from Excel



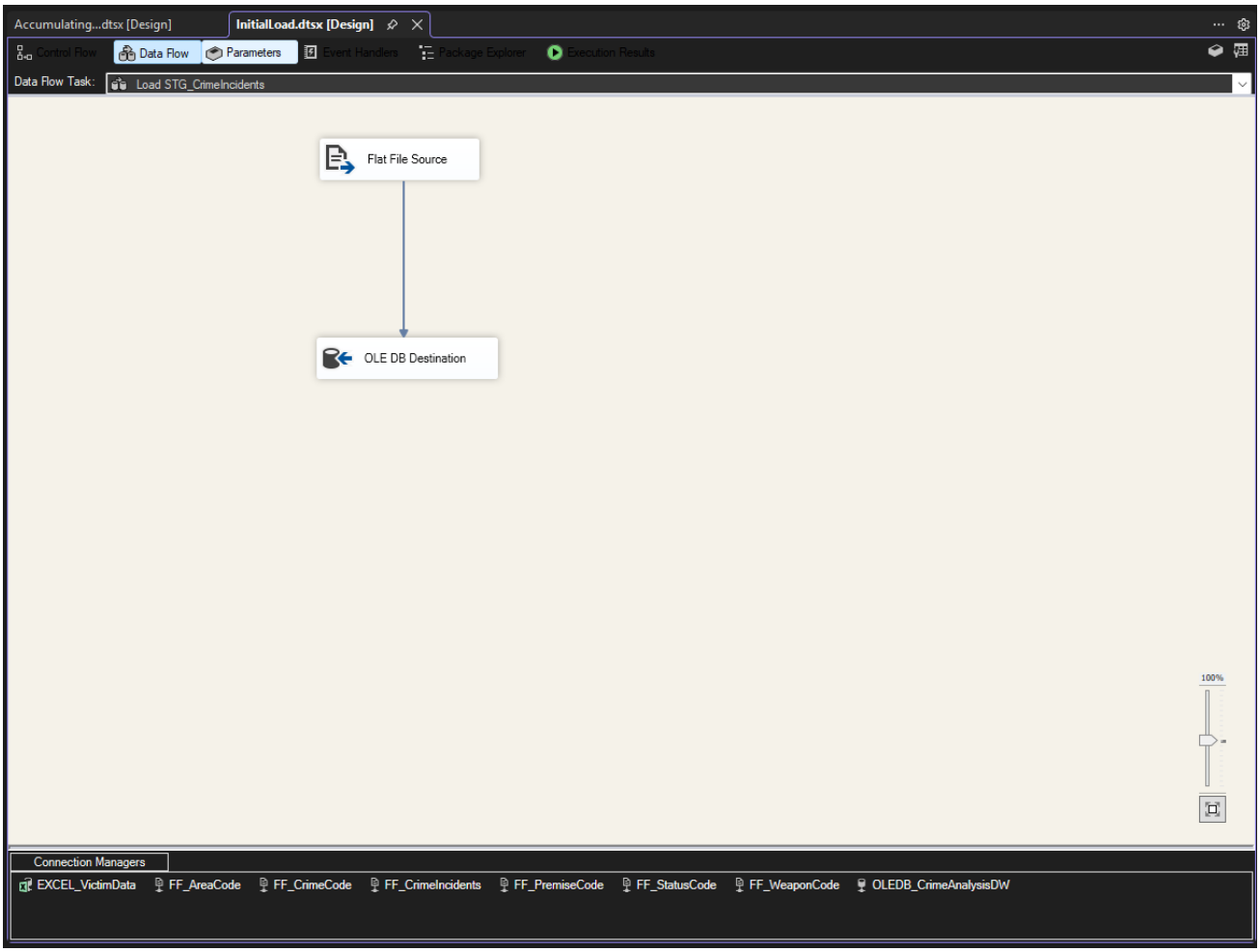
5.2 Full Control Flow – InitialLoad.dtsx

The control flow executes all tasks in strict sequence. Each task must succeed before the next one begins.





5.3 Connection Managers

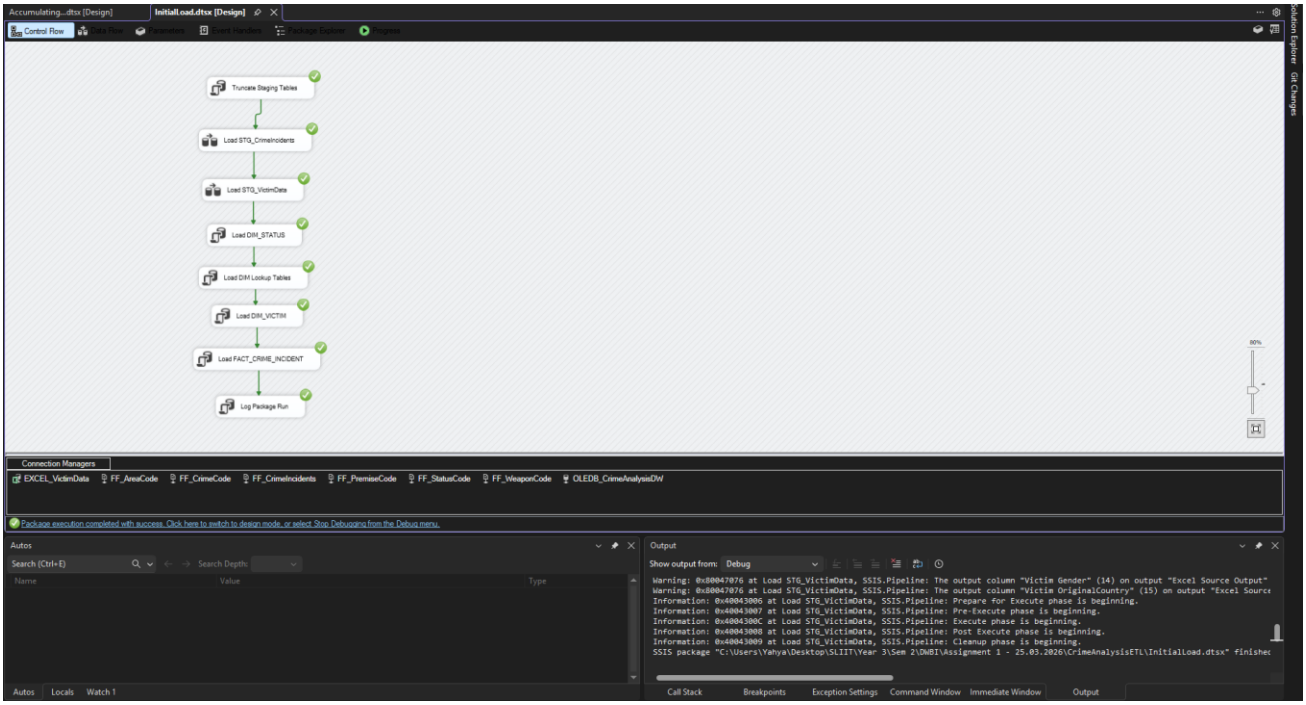


OLEDB_CrimeAnalysisDW	OLE DB	SQL Server 2022 – CrimeAnalysisDW database
EXCEL_VictimData	Excel	VictimData.xlsx – victim demographic data
FF_CrimeIncidents	Flat File	CrimeIncidents.csv – main crime incident records
FF_AreaCode	Flat File	AreaCodeDescription.csv
FF_CrimeCode	Flat File	CrimeCodeDescription.csv
FF_PremiseCode	Flat File	PremiseCodeDescription.csv
FF_StatusCode	Flat File	StatucCodeDescription.csv
FF_WeaponCode	Flat File	WeaponCodeDescription.csv

5.4 ETL Task Descriptions

1	Truncate Staging / Reset	Clears all staging, dimension, and fact tables. Resets IDENTITY counters.
2	Load STG_CrimeIncidents	Flat File Source (CSV) → OLE DB Destination (staging). Loads 50,000 rows.
3	Load STG_VictimData	Excel Source (VictimData.xlsx) → OLE DB Destination. Demonstrates Excel source type.
4	Load DIM_STATUS	Execute SQL Task: inserts 8 case status rows using MERGE.
5	Load DIM Lookup Tables	Execute SQL Task: BULK INSERT 4 lookup CSVs → INSERT into DIM_CRIME, DIM_PREMISE, DIM_WEAPON, DIM_AREA with derived category columns.
6	Load DIM_VICTIM	Execute SQL Task: INSERT into DIM_VICTIM from staging with derived AgeGroup, GenderDescription, DescentDescription. SCD Type 2 structure applied.
7	Load FACT_CRIME_INCIDENT	Execute SQL Task: INSERT 50,000 rows joining staging tables, resolving all surrogate keys, setting accm_txn_create_time = GETDATE().
8	Log Package Run	Execute SQL Task: logs package name, run time and row count to ETL_PackageLog.

5.5 Package Execution – Success



5.6 SSMS Verification After Package 1

```sql
USE CrimeAnalysisDW;

SELECT 'FACT_CRIME_INCIDENT' AS Tbl, COUNT(*) AS Rows FROM FACT_CRIME_INCIDENT
UNION ALL SELECT 'DIM_CRIME',    COUNT(*) FROM DIM_CRIME
UNION ALL SELECT 'DIM_PREMISE',  COUNT(*) FROM DIM_PREMISE
UNION ALL SELECT 'DIM_WEAPON',   COUNT(*) FROM DIM_WEAPON
UNION ALL SELECT 'DIM_AREA',     COUNT(*) FROM DIM_AREA
UNION ALL SELECT 'DIM_STATUS',   COUNT(*) FROM DIM_STATUS
UNION ALL SELECT 'DIM_VICTIM',   COUNT(*) FROM DIM_VICTIM
UNION ALL SELECT 'DIM_DATE',     COUNT(*) FROM DIM_DATE
UNION ALL SELECT 'DIM_TIME',     COUNT(*) FROM DIM_TIME;

-- Check accumulating columns
SELECT TOP 5
    DR_NO,
    accm_txn_create_time,
    accm_txn_complete_time,
    txn_process_time_hours
FROM FACT_CRIME_INCIDENT;
```

100 % 24 0

Results | Messages

	Tbl	Rows
1	FACT_CRIME_INCIDENT	50000
2	DIM_CRIME	140
3	DIM_PREMISE	315
4	DIM_WEAPON	80
5	DIM_AREA	21
6	DIM_STATUS	8
7	DIM_VICTIM	1
8	DIM_DATE	6574
9	DIM_TIME	1440

	DR_NO	accm_txn_create_time	accm_txn_complete_time	txn_process_time_hours
1	230106566	2026-03-28 12:01:20.760	NULL	NULL
2	200809113	2026-03-28 12:01:20.760	NULL	NULL
3	241415803	2026-03-28 12:01:20.760	NULL	NULL
4	201406493	2026-03-28 12:01:20.760	NULL	NULL
5	200805240	2026-03-28 12:01:20.760	NULL	NULL

06. ETL Development – Accumulating Fact Tables

6.1 Accumulating Fact Columns

Three additional columns were added to FACT_CRIME_INCIDENT to implement the accumulating fact table pattern:

accm_txn_create_time	Package 1 – DEFAULT GETDATE()	Set to current system datetime when the fact row is first inserted. Represents when the crime record entered the DW.
accm_txn_complete_time	Package 2 – UPDATE	Initially NULL. Updated by AccumulatingFactUpdate.dtsx when case completion data is received from AccumulatingFactUpdate.csv.
txn_process_time_hours	SQL Server – PERSISTED Computed	Automatically calculated as DATEDIFF(HOUR, accm_txn_create_time, accm_txn_complete_time). Recalculates on every UPDATE to accm_txn_complete_time.

6.2 Update Data File – AccumulatingFactUpdate.csv

A separate CSV file was prepared containing 5,000 case completion records with the following structure:

txn_id	DR_NO of the crime incident – foreign key to FACT_CRIME_INCIDENT.DR_NO
accm_txn_complete_time	Datetime when the case was marked complete (YYYY-MM-DD HH:MM:SS)

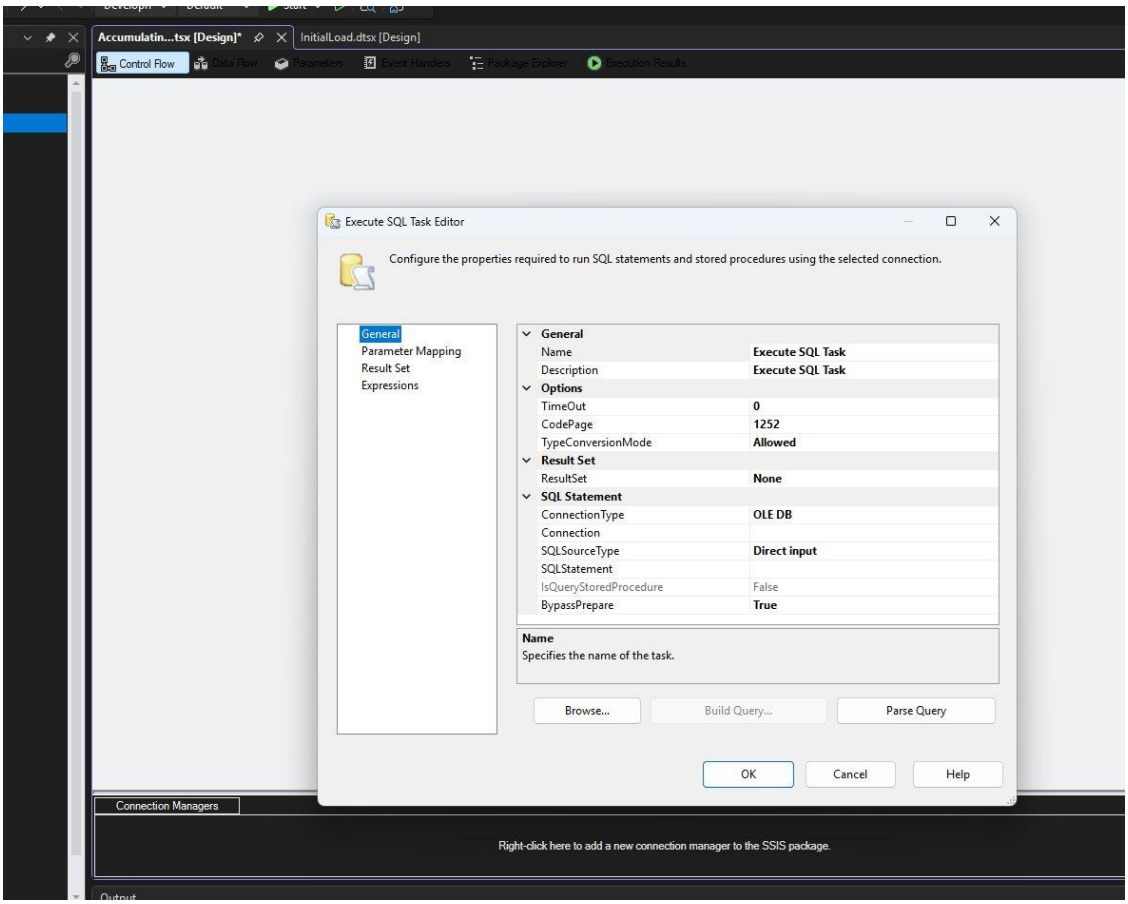
AccumulatingFactUpdate.csv

C: > Users > Yahya > Desktop > SLIIT > Year 3 > Sem 2 > DWBI > Assignment 1 - 25.03.2026 > AccumulatingFactUpdate.csv

```csv
txn_id,accm_txn_complete_time
211111198,2021-11-12 00:00:00
200616094,2020-06-19 07:00:00
210605392,2021-06-07 04:00:00
230119520,2023-01-20 21:00:00
232111853,2023-12-16 02:00:00
221813108,2022-12-18 13:00:00
200808223,2020-08-09 00:00:00
222019121,2022-12-20 06:00:00
220218413,2022-02-20 16:00:00
210315726,2021-03-20 00:00:00
202016524,2020-12-21 06:00:00
201419296,2020-12-24 13:00:00
221519061,2022-12-21 14:00:00
200217524,2020-02-22 08:00:00
220312012,2022-03-13 05:00:00
200120378,2020-01-24 10:00:00
221106436,2022-11-09 04:00:00
230611836,2023-06-13 10:00:00
231319162,2023-12-20 02:00:00
201418391,2020-12-22 03:00:00
202016161,2020-12-19 11:00:00
210710010,2021-07-15 08:00:00
220314533,2022-03-15 23:00:00
242107211,2024-12-11 17:00:00
230512540,2023-05-13 12:00:00
230905531,2023-09-06 17:00:00
220219982,2022-02-22 20:00:00
231216933,2023-12-21 11:00:00
240610035,2024-06-15 06:00:00
211410615,2021-12-11 01:00:00
201104881,2020-11-06 09:00:00
240410864,2024-04-11 07:00:00
220122352,2022-01-23 12:00:00
211005262,2021-10-08 14:00:00
201217543,2020-12-20 05:00:00
240406964,2024-04-09 11:00:00
```

6.3 SSIS Package 2 – AccumulatingFactUpdate.dtsx

A separate lightweight SSIS package was created to process accumulation updates. It uses an Execute SQL Task to run an UPDATE statement joining STG_AccumulatingUpdate to FACT_CRIME_INCIDENT on DR_NO.

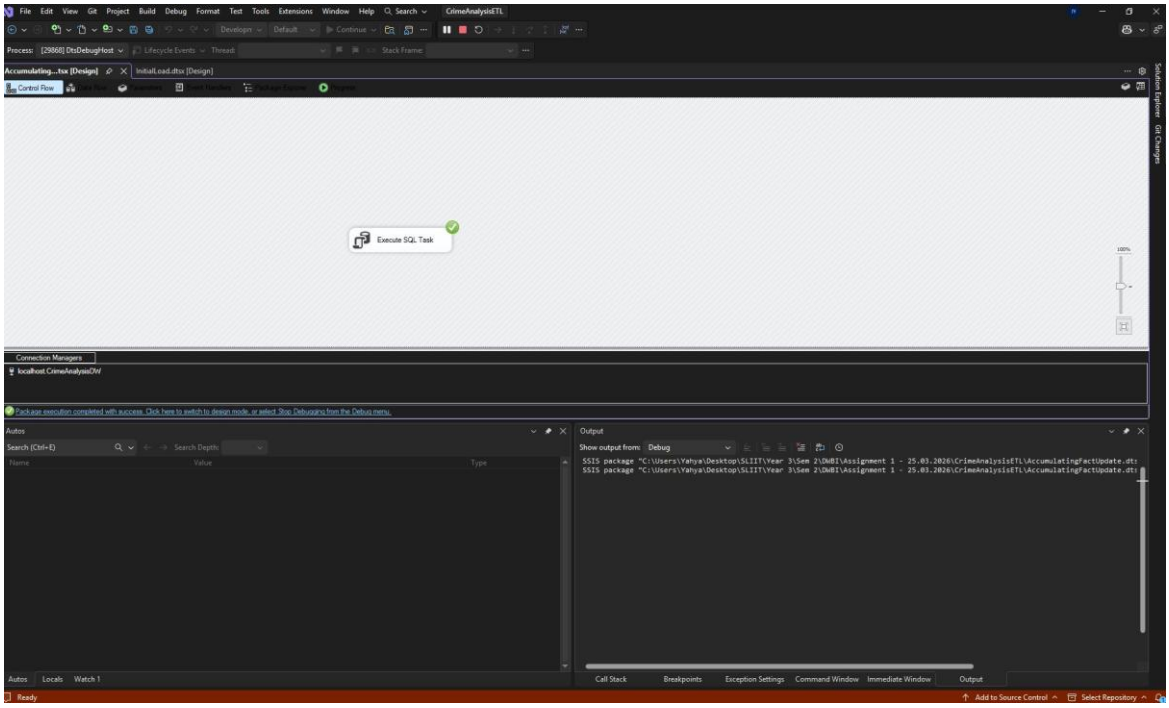


6.4 Update Logic

The Execute SQL Task in Package 2 executes the following logic:

- JOIN STG_AccumulatingUpdate to FACT_CRIME_INCIDENT on DR_NO = txn_id
- SET accm_txn_complete_time = completion timestamp from the CSV
- The PERSISTED computed column txn_process_time_hours automatically recalculates — no additional ETL step required

6.5 Package 2 Execution – Success



6.6 Accumulated Fact Table – Verification

```
773        SUM(CASE WHEN accm_txn_complete_time IS NULL THEN 1 ELSE 0 END) AS Pending
774    FROM FACT_CRIME_INCIDENT;
775
776    -- ETL Log
777    SELECT * FROM ETL_PackageLog ORDER BY RunTime DESC;
```

100 % 24 0

Results | Messages

	DR_NO	accm_txn_create_time	accm_txn_complete_time	txn_process_time_hours
1	200112955	2026-03-28 12:01:20.760	2020-01-15 10:00:00.000	-54338
2	200113347	2026-03-28 12:01:20.760	2020-01-18 00:00:00.000	-54276
3	200120503	2026-03-28 12:01:20.760	2020-01-24 21:00:00.000	-54111
4	200122052	2026-03-28 12:01:20.760	2020-01-26 22:00:00.000	-54062
5	200122121	2026-03-28 12:01:20.760	2020-01-25 17:00:00.000	-54091
6	200206192	2026-03-28 12:01:20.760	2020-02-08 00:00:00.000	-53772
7	200210778	2026-03-28 12:01:20.760	2020-02-12 13:00:00.000	-53663
8	200322213	2026-03-28 12:01:20.760	2020-03-23 07:00:00.000	-52709
9	200409039	2026-03-28 12:01:20.760	2020-04-12 13:00:00.000	-52223
10	200606978	2026-03-28 12:01:20.760	2020-06-10 03:00:00.000	-50817

	TotalFact	Completed	Pending
1	50000	254	49746

	LogID	PackageName	RunTime	RowsLoaded
1	2	AccumulatingFactUpdate	2026-03-28 12:07:57.237	254
2	1	InitialLoad	2026-03-28 12:01:22.660	50000

The query results show:

- TotalFact: 50,000 rows loaded by Package 1
- Completed: 254 rows updated with accm_txn_complete_time by Package 2
- Pending: 49,746 rows awaiting completion (accm_txn_complete_time = NULL)
- txn_process_time_hours: automatically computed by SQL Server for all 254 completed rows
- ETL_PackageLog: both package runs recorded with timestamps and row counts